

Soluciones Certamen 3

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Problem 1

2. A nonrelativistic charged particle of mass m is constrained to move in a horizontal plane, in the field of the harmonic oscillator $U = kr^2/2$ and the constant magnetic field $\mathbf{B}$ perpendicular to the plane. The vector potential $\mathbf{A}$ may be chosen as

$$\mathbf{A} = \frac{1}{2} \mathbf{B} \times \mathbf{r}.$$

Analyze this system in polar coordinates:

Write out the lagrangian, the generalized momenta and construct the hamiltonian

Write out the canonical equations of motion.

Write out the equation of Hamilton-Jacobi and integrate it using separation of variables.

Note: *The goal of this part is to test your knowledge of the Hamilton-Jacobi formalism, for this reason solution with any other method will not be taken into account.*

Lagrangian (see Lecture 12; in polar coordinates $\mathbf{v} = \dot{r}\hat{\mathbf{r}} + r\dot{\varphi}\hat{\boldsymbol{\varphi}} + \dot{z}\hat{\mathbf{z}}$):

$$L = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\varphi}^2) - \frac{kr^2}{2} + q \underbrace{\frac{Br^2}{2}}_{\mathbf{v} \cdot \mathbf{A}} \dot{\varphi}, \quad \mathbf{A} = \frac{1}{2} \mathbf{B} \times \mathbf{r} = \frac{Br}{2} \hat{\boldsymbol{\varphi}}$$

► Generalized momenta:

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r}, \quad p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = m r^2 \dot{\varphi} + q \frac{Br^2}{2},$$

—no dependence on φ , cyclic variable, $p_\varphi = \text{const}$

► Construct hamiltonian using Legendre transform:

$$H \equiv \sum_a p_a \dot{q}_a - L = p_r \dot{r} + p_\varphi \dot{\varphi} - L = \frac{p_r^2}{2m} + \frac{\left(p_\varphi - q \frac{Br^2}{2}\right)^2}{2mr^2} + \frac{kr^2}{2}$$

► Canonical equations:

$$\dot{p}_r = -\frac{\partial H}{\partial r} = -\frac{\left(q \frac{B}{2}\right)^2 r^4 - p_\varphi^2}{mr^3} - kr, \quad \dot{p}_\varphi = -\frac{\partial H}{\partial \varphi} = 0,$$

$$\dot{r} = \frac{\partial H}{\partial p_r} = \frac{p_r}{m}, \quad \dot{\varphi} = \frac{\partial H}{\partial p_\varphi} = \frac{\left(p_\varphi - q \frac{Br^2}{2}\right)}{m}.$$

Problem 1 - Hamilton-Jacobi formalism

$$\frac{\partial S}{\partial t} + H\left(p_a = \frac{\partial S}{\partial q_a}, q_a\right) = 0$$

- Energy is conserved since H does not depend on time
- Apply separation of variables: effectively assume that S has a form

$$S = -Et + a(r) + b(\varphi)$$

$$E = \frac{(a'(r))^2}{2m} + \frac{\left(b'(\varphi) - q\frac{Br^2}{2}\right)^2}{2mr^2} + \frac{kr^2}{2}$$
$$2mE = (a'(r))^2 + \frac{1}{r^2} \left(b'(\varphi) - q\frac{Br^2}{2}\right)^2 + kmr^2$$

We can see that φ -dependence appears only in $b'(\varphi)$, so must conclude that

$$b'(\varphi) = p_\varphi = \text{const}$$

Problem 1

$$\left(\frac{da}{dr}\right)^2 = 2mE - kmr^2 - \frac{1}{r^2} \left(p_\varphi - q\frac{Br^2}{2}\right)^2$$

$$\begin{aligned}\Rightarrow a(r) &= \int^r dr \sqrt{2mE - kmr^2 - \frac{1}{r^2} \left(p_\varphi - q\frac{Br^2}{2}\right)^2} \\ &= \int^r dr \sqrt{2m \left(E + \frac{qB}{2m}p_\varphi\right) - m \left(k + \frac{q^2 B^2}{4m}\right) r^2 - \frac{p_\varphi^2}{r^2}}\end{aligned}$$

► Same integral as for plane harmonic oscillator in polar coordinates

- but with “adjusted” energy $\tilde{E} = E + \frac{qB}{2m}p_\varphi$ and elasticity $\tilde{k} = k + \frac{q^2 B^2}{4m}$
- i.e. the magnetic field just changes parameters

(Mathematica returns the result in terms of elementary functions but too lengthy)

$$\text{action : } S = -Et + a(r) + p_\varphi \varphi$$

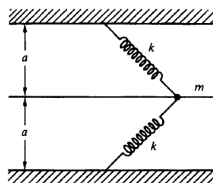
► Now, after we got S , we can extract trajectories from

$$\frac{\partial S}{\partial E} = \text{const} = -t_0, \quad \frac{\partial S}{\partial p_\varphi} = \text{const} = -\varphi_0.$$

Problem 2

2. (8 points)

A particle of mass m is constrained to move along a horizontal straight rail without friction. Two springs of elasticity k are attached to the spring as shown in the Figure. The lengths of unstretched springs $b < a$. The system might oscillate near its position of equilibrium.



(6 point) Assuming that $\Delta H = H - H_0$ is a small perturbation, and using canonical perturbation theory, calculate:

(3 points) The first nonvanishing correction to the trajectory of the particle

(3 points) The first nonvanishing correction to the frequency of oscillations.

Problem 2 (solution)

The proper lengths of unstretched springs $b < a$

⇒ The minimal potential energy (equilibrium) position corresponds to the mass located between the two support points of the springs

If x is horizontal deviation of the mass from its equilibrium, then the elongation

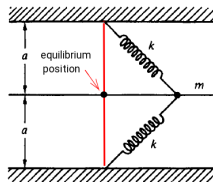
$$\ell_{1,2} = \sqrt{x^2 + a^2} - b$$

Potential energy:

$$U(x) \approx 2 \times \frac{k\ell^2}{2} = k \left(\sqrt{x^2 + a^2} - b \right)^2$$

$$\approx k \left(a + \frac{x^2}{2a} - \frac{x^4}{8a^3} + \mathcal{O}(x^6) - b \right)^2$$

$$\approx k \left[(a - b)^2 + x^2 \frac{a - b}{a} + \frac{b}{4a^3} x^4 + \mathcal{O}(x^6) \right]$$



Problem 2 (solution)

Hamiltonian of the system

$$H = \frac{p^2}{2m} + U(x) = H_0 + H_1,$$

$$H_0 = \frac{p^2}{2m} + kx^2 \frac{a-b}{a}, \quad H_1 \approx \frac{kb}{4a^3}x^4 + \mathcal{O}(x^6)$$

For H_0 we know that solutions correspond to oscillations with eigenfrequency

$$\omega_0 = \sqrt{\frac{2k}{m} \frac{a-b}{a}}$$

For H_1 we have classical anharmonic oscillator with $\sim \lambda x^4$ which we analyzed in Lecture 18, pages 18-21. We only need to adjust coefficients

$$k \rightarrow 2k \frac{a-b}{a}, \quad \lambda \rightarrow \frac{kb}{a^3}$$

Problem 2 (solution)

► Solution of 0-order problem:

$$q_0(t) = \sqrt{\frac{\alpha a}{k(a-b)}} \sin(\omega t + \beta), \quad \omega \equiv \sqrt{\frac{2k}{m} \frac{a-b}{a}}$$

● Perturbation hamiltonian:

$$\Delta H = \frac{\lambda q_0^4}{4} = \frac{\alpha^2 b}{4ak(a-b)^2} \sin^4(\omega t + \beta)$$

● Canonical equations in terms of α, β :

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta} = -\frac{\alpha^2 b}{ak(a-b)^2} \sin^3(\omega t + \beta) \cos(\omega t + \beta)$$

$$\dot{\beta} = \frac{\partial \Delta H}{\partial \alpha} = \frac{\alpha b}{2ak(a-b)^2} \sin^4(\omega t + \beta)$$

● No known analytical solutions

—Develop perturbation theory

Problem 2 (solution)

$$\dot{\alpha} = -\frac{\partial \Delta H}{\partial \beta} = -\frac{\alpha^2 b \sin^3(\omega t + \beta) \cos(\omega t + \beta)}{ak(a-b)^2}, \quad \dot{\beta} = \frac{\partial \Delta H}{\partial \alpha} = \frac{\alpha b \sin^4(\omega t + \beta)}{2ak(a-b)^2}$$

—Develop perturbation theory:

$$\alpha \approx \alpha_0 + \alpha_1(t), \quad |\alpha_1| \ll |\alpha_0|, \quad \beta \approx \beta_0 + \beta_1(t), \quad |\beta_1| \ll |\beta_0|$$

$$\dot{\alpha}_1 = -\frac{\alpha_0^2 b \sin^3(\omega t + \beta_0) \cos(\omega t + \beta_0)}{ak(a-b)^2} \Rightarrow \alpha_1(t) = -\frac{\alpha_0^2 b \sin^4(\omega t + \beta_0)}{ak(a-b)^2 \omega}$$

$$\dot{\beta}_1 = \frac{\alpha b \sin^4(\omega t + \beta)}{2ak(a-b)^2}$$

$$\begin{aligned} \Rightarrow \beta_1(t) = & \frac{3\alpha_0 b}{16 ka(a-b)^2 \omega^2} (\omega t + \beta_0) + \\ & + \frac{\alpha_0 a^2 b}{64 ka(a-b)^2 \omega^2} \frac{\sin(4(\omega t + \beta_0)) - 8 \sin(2(\omega t + \beta_0))}{32\omega} \end{aligned}$$

Problem 2 (solution)

Full solution: $\alpha_1(t)$ affects amplitude of oscillations

$\beta_1(t)$ affects **phase** and **frequency** of oscillation:

$$\begin{aligned}\sin(\omega t + \beta) &= \sin(\omega t + \beta_0 + \beta_1) = \\ &= \sin \left[\left(\omega + \frac{3\alpha_0 b}{16 k a (a-b)^2 \omega} \right) t + \beta_0 \left(1 + \frac{3\alpha_0 b}{16 k a (a-b)^2 \omega^2} \right) + \right. \\ &\quad \left. + \frac{\alpha_0 a^2 b}{64 k a (a-b)^2 \omega^2} \frac{\sin(4(\omega t + \beta_0)) - 8 \sin(2(\omega t + \beta_0))}{32\omega} \right]\end{aligned}$$

Final result for change of frequency:

$$\delta\omega = \frac{3b\alpha_0}{16ak(a-b)^2\omega_0} = \frac{3b\alpha_0}{16ka^{3/2}(a-b)^{3/2}} \sqrt{\frac{2k}{m}}.$$

So far we used canonical perturbation theory which depends on time

We could also use a canonical perturbation theory in time-independent form (in terms of action-angle variables)

Problem 2 (solution)

Derivation in a canonical perturbation theory in time-independent form (in terms of action-angle variables)

Analyzed in Lecture 19 for $H_1 \sim \lambda q^3$ perturbation, need to repeat now for our potential $\sim \lambda x^4$

We found the analytic expression for $x_0(t)$ (trajectory for H_0) in terms of action-angle variables in Lecture 19

$$H_0 = \omega_0 I_0 = E, \quad w_0 \equiv \frac{\partial S_0}{\partial I_0} = \dots = \arctan \left(\frac{x}{\sqrt{x_{\max}^2 - x^2}} \right)$$

$$x_{(0)}(t) = \sqrt{\frac{I_0}{m\omega_0}} \sin w_0, \quad x_{\max} = \sqrt{\frac{I}{m\omega_0}} = \sqrt{\frac{E}{m\omega_0^2}}$$

Now substitute this into H_1

$$\Rightarrow H_1 = \frac{b}{4a^3} \left(\frac{I_0}{m\omega_0} \right)^2 \sin^4 w_0$$

Problem 2 (solution)

Reconstruct $H(I)$ from $H_0(I_0)$ and H_1 using general expressions from Lecture 19:

$$H_0 = \omega_0 I_0 = E, \quad H_1 = \frac{b}{4a^3} \left(\frac{I_0}{m\omega_0} \right)^2 \sin^4 w_0$$

$$H(I) = h_0(I) + h_1(I) + \dots$$

$$h_0(I) = H_0(I),$$

$$\begin{aligned} h_1(I) &= \overline{H_1(\mathbf{w}_0, I)} = \frac{1}{2\pi} \oint dw_0 H_1(\mathbf{w}_0, I) \\ &= \frac{b}{4a^3} \left(\frac{I_0}{m\omega_0} \right)^2 \frac{3}{8} = \frac{3b}{32a^3} \left(\frac{I_0}{m\omega_0} \right)^2 \neq 0 \end{aligned}$$

where we used

$$\int_0^{2\pi} dw \sin^4 w = \frac{3}{8} 2\pi \neq 0$$

Problem 2 (solution)

Reconstruct $H(I)$ from $H_0(I_0)$ and H_1 using general expressions from Lecture 19:

$$H_0 = \omega_0 I_0 = E, \quad H_1 = \frac{b}{4a^3} \left(\frac{I_0}{m\omega_0} \right)^2 \sin^4 w_0$$

$$H(I) = h_0(I) + h_1(I) + \dots$$

$$\Rightarrow H(I) = I\omega_0 + \frac{3b}{32a^3} \left(\frac{I_0}{m\omega_0} \right)^2$$

Taking the derivative, may get

$$\omega \equiv \dot{w} = \frac{\partial \bar{H}}{\partial I} = \omega_0 + \delta\omega, \quad \delta\omega = \frac{\partial h_1(I)}{\partial I_0} = \frac{3b}{16a^3} \left(\frac{I_0}{m^2\omega_0^2} \right).$$

If we now use $E = I_0\omega_0 = \alpha_0$, we'll get the same result as in time-dependent canonical theory

$$\delta\omega = \frac{3b\alpha_0}{2}.$$